

Final group pest risk analysis for mealybugs and the viruses they transmit on fresh fruit, vegetable, cut-flower and foliage imports

The Department of Agriculture and Water Resources has completed its [group pest risk analysis \(PRA\)](#) for for mealybugs and the viruses they transmit on fresh fruit, vegetables, cut-flowers and foliage imports.

This is the second Group PRA to be finalised—the first Group PRA was for thrips and orthotospoviruses. This second Group PRA considers the biosecurity risk posed by all members of the Pseudococcidae, Putoidae and Rhizoecidae families, commonly referred to as mealybugs, which in total comprise about 2,300 described species. In addition the Group PRA considers all viruses transmitted by mealybugs that are (or are likely to be) associated with fresh fruit, vegetables, cut flowers or foliage imported into Australia as commercial consignments. We followed 3 principal steps when conducting this group PRA:

- Our experts conducted a review of scientific knowledge relevant to the group of pests and prepared a PRA.
- We released the draft group PRA for a 60-day public consultation period on 28 September 2018 via [Biosecurity Advice 2018-24](#). The draft report identified the biosecurity risks and risk management measures required to achieve Australia's appropriate level of protection (ALOP).
- We finalised the group PRA, taking into consideration all stakeholder submissions, and released the [final group pest risk analysis mealybugs and the viruses they transmit on fresh fruit, vegetables, cut-flowers and foliage imports](#) on 31 January 2019 via [Biosecurity Advice 2019-01](#).

This group pest risk analysis is funded through the Australian Government's [Agricultural Competitiveness White Paper](#) to strengthen biosecurity surveillance and analysis.

Final Report Summary

The Department of Agriculture and Water Resources is improving the effectiveness and consistency of the Pest Risk Analysis (PRA) process. A key step in this improvement is the development of the Group PRA, which considers the biosecurity risk posed by a group of pests across numerous import pathways. It applies the significant body of available scientific knowledge, including pest interception data and previous PRAs, to provide an overarching analysis of the risks posed by the group.

The International Plant Protection Convention (IPPC) defines PRA as 'the process of evaluating biological or other scientific and economic evidence to determine whether an organism is a pest, whether it should be regulated, and the strength of any phytosanitary measures to be taken against it' (FAO 2017b). International Standard for Phytosanitary Measures (ISPM) 2: Framework for pest risk analysis (FAO 2016a), states that 'Specific organisms may ... be analysed individually, or in groups where individual species share common biological characteristics.' This is the basis for the Group PRA,

in which organisms are grouped if they share common biological characteristics, and as a result also have similar likelihoods of entry, establishment and spread and comparable consequences—thus posing a similar level of biosecurity risk.

Undertaking and utilising PRAs on groups of pests that share common biological characteristics provides significant opportunities to improve effectiveness and consistency of commodity-based PRAs with which those pests are also associated and to maintain a high level of biosecurity protection against new and emerging risks. The group approach to PRA was initiated by the department to take advantage of these opportunities. It is a ‘building block’ that can be used to review existing trade pathways, and can also be applied to prospective pathways for which a specific PRA is required.

If a Group PRA is used to review existing or new trade pathways there may be no need to undertake further detailed PRAs on these pests—if the trade-dependent factors relating to the likelihood of entry on specific pathways have been verified, the Group PRA can be applied.

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Mealybugs and the viruses they transmit can have consequences across a range of crops by reducing yield, quality and marketability.

This Group PRA identifies and analyses the key quarantine pests of biosecurity importance to Australia. It is built on a foundation of 19 years of PRAs undertaken by the department, all of which were subjected to robust scientific analyses and extensive stakeholder consultation. These pest risk assessments showed marked consistency in the level of biosecurity risk posed by mealybugs relative to the appropriate level of protection (ALOP) for Australia. They also indicated that certain mealybug species are associated with a broad range of plant commodities from many countries.

This report’s conclusions have been validated with available scientific evidence including 30 years of interception data collected at Australia’s borders, similar interception records available from other countries, and an extensive literature review. The report includes significant pests that have been recognised internationally, by Australian industry, and those identified by states and territories as regional pests for Australia.

Selection criteria were used to identify mealybug species with potential biosecurity importance for Australia. One hundred and sixty-nine species were confirmed as quarantine pests for Australia. The Group PRA also identified nine viruses transmitted by mealybugs that are quarantine pests for Australia.

Mealybug quarantine pests were estimated to have an ‘indicative’ unrestricted risk estimate of ‘Low’ which does not achieve the appropriate level of protection (ALOP) for Australia. This risk estimate is regarded as ‘indicative’ because the likelihood of entry (importation and distribution) can be influenced

by a range of pathway-specific factors (such as the commodity, seasonal considerations, or the incidence of mealybugs in specific export production areas), and must be verified on a case-by-case basis. In some cases the likelihood of entry may need to be adjusted to take account of these factors. In order to achieve an appropriate level of protection for Australia, measures will be required for quarantine mealybugs when the unrestricted risk estimate of 'Low' has been confirmed for a specific plant import pathway.

In contrast, the viruses of biosecurity concern transmitted by mealybugs were estimated to have an 'indicative' unrestricted risk estimate of 'Very Low' for the plant import pathway, which achieves the ALOP for Australia. This is because mealybugs can only transmit viruses for a short period of time (semi-persistent transmission) and these viruses also have a limited host range compared to their mealybug vectors. These biological factors significantly limit the likelihood that mealybugs present on imported fresh fruit, vegetable, cut-flowers and foliage will be able to transmit exotic viruses to a host plant in Australia. Therefore no additional measures are required for these viruses transmitted by mealybugs on the plant import pathway.

Imported commodities will be regulated if they are infested with mealybug quarantine pests to reduce the risk of establishment of these organisms in Australia. Regulation will be in accordance with this PRA and any other relevant commodity-based PRAs.

Phytosanitary measures will also be required if the indicative unrestricted risk estimate is verified for a specific plant import pathway and the ALOP for Australia is not achieved.

The Group PRA identifies measures for mealybug quarantine pests, and alternative risk management options that may be considered on a case-by-case basis when developing new import conditions for specific commodities, or when reviewing existing import conditions for commodities that are currently traded. These measures are consistent with long-standing established import requirements for mealybug quarantine pests.

Measures are applied to ensure that goods in consignments are free from mealybug quarantine pests. Verification measures, such as inspection, are required to provide assurance that Australia's import conditions have been met and the appropriate level of protection achieved. Additional operational procedures may be required on a case-by-case basis for specific plant import pathways, such as a system of traceability, registration of packing house and treatment providers and auditing of procedures, packaging and labelling requirements and specific conditions for storage and movement.

Imported goods that are frequently found to be infested with mealybug quarantine pests may be subject to mandatory treatment, which may be required pre-export rather than as a remedial action on arrival.

Next steps

We will use this group PRA when reviewing existing import conditions, or when developing new import conditions, for specific commodities when thrips are identified.

New scientific information

Scientific information can be provided to us at any time, including after a risk analysis has been completed. We will consider the information provided and review the analysis.

More information

For more information, you can [email us](#) or phone +61 2 6272 5094.

Source: <https://www.agriculture.gov.au/biosecurity-trade/policy/risk-analysis/plant/group-pest-risk-analyses/mealybugs>